the Almost Perfect Generative Michelin Star

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of the software performing arts research krew.

beauty

Science makes sense of the beautiful mysteries of the world. It proposes ways to understand the invisible, the intangible, the untestable. On this journey, many scientists wish to measure phenomena that are not directly measurable. In design science, this process is referred to as operationalization. Intuitively, the operationalization process is similar to what literature or cinema critics do when they assign a rating to a movie or novel review [7, 6].

When exploring the intangible, we collect evidence, metrics, impressions and make sense of it. For example, software easter eggs researchers look for easters in the most improbable corners of the internet, count them, count the lines of code, measure the length of execution traces, and eventually collect the impressions from easter egg hunters. Based on sound and quiet processes, these elements are turned into an operationalized assessment; “that’s a four-star easter egg!”

In this search for operationalization, painting, dance, music, cinema, sculpture, sports, or food experts have created a vast and diverse language of graphical cues to represent ratings. From various forms of stars and happy faces to more domain-specific icons such as footballs, sailing boats or tires. Today these cues can be found in a variety of media and pop culture publication, from the regular thumbs-up 👍, star ⭐, clap 👏, or smiley face 😃 to more emotional heart ❤️, heart eyes 😍 or even star struck 🤩. All these cues can be used to rank and categorize objects and concepts that are non categorical by nature, such as a great movie ⭐⭐⭐, a good software easter egg ⭐, or a world-class package manager to host, distribute and secure Minecraft mods 🥇.

Good news for beautiful science [2], a vast majority of these icons are natively available for the scientific literature thanks to LuaLaTeXand the emoji package [8].

Yet, there is one shortcoming for truly beautiful, sophisticated, gourmet, rankings in the scientific literature. The iconic Michelin star is not available in unicode, leaving scientists, researchers, students, and scientific journalists totally helpless when it comes to ranking with class. Michelin stars inspire [1, 3] and we wish to let scientists enjoy it as much as the anonymous Michelin jurors do when they sit at a new table.

In this work, we contribute to the enchantment of scientific literature with a high resolution, scalable, almost perfect Michelin star, in Tikz [5].

uncanny

This work is prepared and edited in the ooooold tradition of humor in science. Sitting on the shoulders of the Monty Python and all IgNobel laureates, we wish to share our serious and bustling journey to the center of TikZ, searching for stars. Prompts and floating points numbers we found, as well as mysterious glyphs of ancient continuous integration pipelines. With no compass app, other than our sense of beauty and our appetite, we searched and digged. Rewarded we were, on the day the paper compiled [4]. This tale is nothing short of code and red ({RGB}{206,93,75} to be precise, because when you like colors and code, well you want to be precise and because humor does in no way prevent us from liking number and curly brackets) and we hope you will appreciate it.

astronomy

Our tale begins in the glorious era where AI confidently answers questions we did not even think to ask. Confident and impatient we naturally turn to the all-powerful Claude Sonnet 4.6 and ask for TikZ code to draw a Michelin star. 😎. Claude’s legendary swiftness graciously generates a piece of TikZ code in less time than we need to get to the first bite of a Michelin dinner. Feverish, we can’t eat. We need to execute the code (Listing 3). It looks complicated, packed with enough mysterious numbers to convince us it must be correct. 🥴 We compile it, we execute it, and ...... we get a red, strange looking, glitchy, almost tired Michelin star. the `buggy Michelin star`.

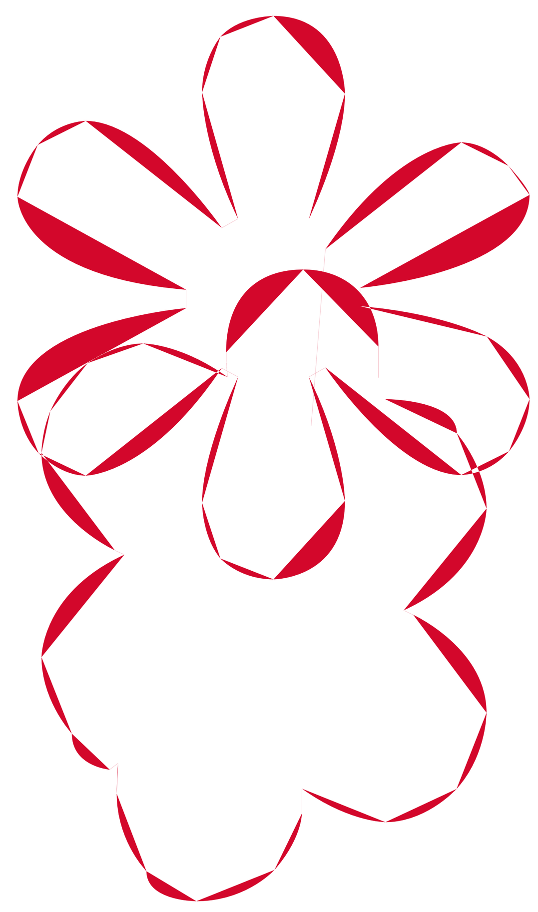
We appreciate its abstract beauty, as well as Claude’s diligent efforts. But our faith in beautiful science fuels our ambition to create a fresh, delicate, sophisticated Michelin star. So, in a moment of nostalgic wisdom, we discover a Python package called svg2tikz with good, old fashioned human-written code. 😍. Feverish again, we search the internet for an standard vector graphics version of the Michelin star. As soon as we find it, we feed it into the converter. We execute and done. We get another impressively complex TikZ code, which we present in Listing 1. To our surprise, this version generates the `glitchy Michelin star`, a star that is closer to the real thing, 👏 still a little glitchy. At this point, we decide the `glitchy Michelin star` has potential. So, we return to Claude Sonnet 4.6, present it with this improved but imperfect TikZ code, and ask it to fix the glitches. And....... tada! 🎉🎉🎉. We have it! The spectacular TikZ code presented in Listing 2 that delivers a smooth, red star full of savour and flowers, full of dreams of dishes that cannot exist and rare porcelin, a true appetizer, an enhancer of scientific discoveries, a reward after any long-running experiment and uncanny major revision for reviewer 2. The One. The Only. The `almost perfect Michelin star`.

```
\def \globalscale {0.0200000}
\newcommand\scoremichelin{
    \begin{tikzpicture}[y=1cm, x=1cm, yscale=\globalscale,xscale=\
        globalscale, every node/.append style={scale=\globalscale}, inner
         sep=0pt, outer sep=0pt]
      \path[fill=red] (16.51, -1.8256) -- (16.51, -0.5292) .. controls
        (16.51, 0.7761) and (16.0955, 1.8521) .. (15.2665, 2.6988) ..
        controls (10.7421, 3.9688) and (9.6661, 3.5366) .. (8.89,
        2.6723) -- (7.7258, -1.1906) .. controls (7.7258, -1.2612) and
        (7.7347, -1.3229) .. (7.7523, -1.3758) -- (7.7788, -1.8256) ..
        controls (6.4735, -0.8908) and (5.3005, -0.4233) .. (4.2598,
        -0.4233) .. controls (3.4484, -0.4233) and (2.6723, -0.6967) ..
         (1.9315, -1.2435) .. controls (1.2259, -1.7551) and (0.7056,
        -2.4253) .. (0.3704, -3.2544) .. controls (0.1235, -3.8365) and
         (0.0, -4.4185) .. (0.0, -5.0006) .. controls (0.0, -6.7822)
        and (1.0231, -8.1403) .. (3.0692, -9.0752) -- (3.466, -9.2604)
        .. controls (1.1553, -10.3188) and (0.0, -11.7475) .. (0.0,
        -13.5467) .. controls (0.0, -14.7638) and (0.4233, -15.8309) ..
         (1.27, -16.7481) .. controls (5.3446, -18.124) and (6.3941,
        -17.7447) .. (7.4348, -16.9863) -- (7.7788, -16.7217) ..
        controls (7.7435, -17.0744) and (7.7258, -17.4978) .. (7.7258,
        -17.9917) .. controls (7.7258, -19.3146) and (8.1403, -20.395)
        .. (8.9694, -21.2328) .. controls (13.5114, -22.4896) and
        (14.5918, -22.053) .. (15.3591, -21.1799) -- (16.51, -16.7217)
        .. controls (17.8153, -17.6565) and (18.9794, -18.124) ..
        (20.0025, -18.124) .. controls (21.149, -18.124) and (22.1412,
        -17.6609) .. (22.9791, -16.7349) .. controls (24.2358,
        -11.7475) and (23.2128, -10.3805) .. (21.1667, -9.4456) --
        (20.7698, -9.2604) .. controls (23.0805, -8.2021) and (24.2358,
         -6.7822) .. (24.2358, -5.0006) .. controls (24.2358, -3.8541)
        and (23.8213, -2.8046) .. (22.9923, -1.8521) -- cycle(14.2875,
        -6.2971) .. controls (16.3336, -3.3161) and (18.2298, -1.8256)
        .. (19.976, -1.8256) .. controls (20.6992, -1.8256) and
        (21.3563, -2.1519) .. (21.9472, -2.8046) .. controls (22.8335,
        -7.1878) and (20.4699, -8.4843) .. (15.7427, -8.89) --
        (15.7427, -9.6573) .. controls (18.0887, -9.8513) and (19.857,
        -10.2747) .. (21.0476, -10.9273) .. controls (22.8335,
        -14.3404) and (22.5469, -15.068) .. (21.9736, -15.7295) ..
        controls (18.1945, -16.7217) and (16.2983, -15.2224) ..
        (14.2875, -12.2238) -- (13.5996, -12.6206) .. controls (14.605,
         -14.6667) and (15.1077, -16.3953) .. (15.1077, -17.8065) ..
        controls (15.1077, -19.9937) and (14.1111, -21.0873) ..
        (12.1179, -21.0873) .. controls (11.1654, -21.0873) and
        (10.4246, -20.7963) .. (9.8954, -20.2142) .. controls (9.3839,
        -19.5968) and (9.1281, -18.8207) .. (9.1281, -17.8858) ..
        controls (9.1281, -16.4747) and (9.6308, -14.7197) .. (10.6363,
         -12.6206) -- (9.9483, -12.2238) .. controls (7.9375, -15.2224)
         and (6.0413, -16.7217) .. (4.2598, -16.7217) .. controls
        (3.5366, -16.7217) and (2.8795, -16.4086) .. (2.2886, -15.7824)
         .. controls (1.4023, -11.3594) and (3.7571, -10.063) ..
        (8.4667, -9.6573) -- (8.4667, -8.89) .. controls (3.7571,
        -8.5019) and (1.4023, -7.2055) .. (1.4023, -5.0006) .. controls
         (1.4023, -4.2245) and (1.6845, -3.5013) .. (2.249, -2.831) ..
        controls (6.006, -1.8256) and (7.9022, -3.3161) .. (9.9483,
        -6.2971) -- (10.6363, -5.9267) .. controls (9.6308, -3.8629)
        and (9.1281, -2.099) .. (9.1281, -0.635) .. controls (9.1281,
        0.2999) and (9.3839, 1.076) .. (9.8954, 1.6933) .. controls
        (10.4246, 2.2754) and (11.1654, 2.5665) .. (12.1179, 2.5665) ..
         controls (14.1111, 2.5665) and (15.1077, 1.4817) .. (15.1077,
        -0.6879) .. controls (15.1077, -2.1167) and (14.605, -3.8629)
        .. (13.5996, -5.9267) -- cycle;
    \end{tikzpicture}
}
```

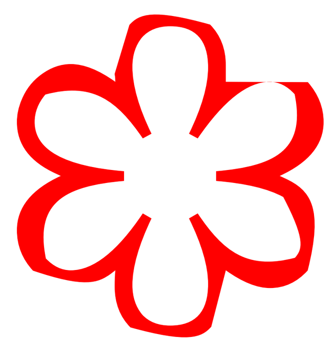
Listing 1: The glitchy TikZ code towards a Michelin star 👍

```
\def \globalscale {0.3}
\newcommand\michelin{
    \begin{tikzpicture}[y=1cm, x=1cm, yscale=\globalscale,xscale=\
        globalscale, every node/.append style={scale=\globalscale}, inner
         sep=0pt, outer sep=0pt]
      % Outer path - stroked with consistent line width
      \path[draw=red, line width=8, line cap=round, line join=round,
         fill=none]
        (12.118, 2.567)
        .. controls (10.166, 2.567) and (9.384, 1.693) ..
        (8.895, 0.275)
        .. controls (8.590, -0.900) and (8.467, -2.540) ..
        (9.948, -6.297)
        .. controls (7.902, -3.316) and (6.006, -1.826) ..
        (4.260, -1.826)
        .. controls (2.514, -1.826) and (1.402, -3.316) ..
        (1.402, -5.001)
        .. controls (1.402, -7.206) and (3.757, -8.502) ..
        (8.467, -8.890)
        -- (8.467, -9.657)
        .. controls (3.757, -10.063) and (1.402, -11.359) ..
        (1.402, -13.547)
        .. controls (1.402, -15.232) and (2.514, -16.722) ..
        (4.260, -16.722)
       .. controls (6.006, -16.722) and (7.902, -15.232) ..
       (9.948, -12.224)
        .. controls (8.467, -15.981) and (8.590, -17.621) ..
        (8.895, -18.796)
        .. controls (9.384, -20.214) and (10.166, -21.087) ..
        (12.118, -21.087)
        .. controls (14.070, -21.087) and (14.852, -20.214) ..
        (15.341, -18.796)
        .. controls (15.646, -17.621) and (15.769, -15.981) ..
        (14.288, -12.224)
        .. controls (16.334, -15.232) and (18.230, -16.722) ..
        (19.976, -16.722)
        .. controls (21.722, -16.722) and (22.834, -15.232) ..
        (22.834, -13.547)
        .. controls (22.834, -11.359) and (20.479, -10.063) ..
        (15.743, -9.657)
        -- (15.743, -8.890)
        .. controls (20.479, -8.502) and (22.834, -7.206) ..
        (22.834, -5.001)
        .. controls (22.834, -3.316) and (21.722, -1.826) ..
        (19.976, -1.826)
        .. controls (18.230, -1.826) and (16.334, -3.316) ..
        (14.288, -6.297)
        .. controls (15.769, -2.540) and (15.646, -0.900) ..
        (15.341, 0.275)
        .. controls (14.852, 1.693) and (14.070, 2.567) ..
        (12.118, 2.567)
        -- cycle;
    \end{tikzpicture}
}
```

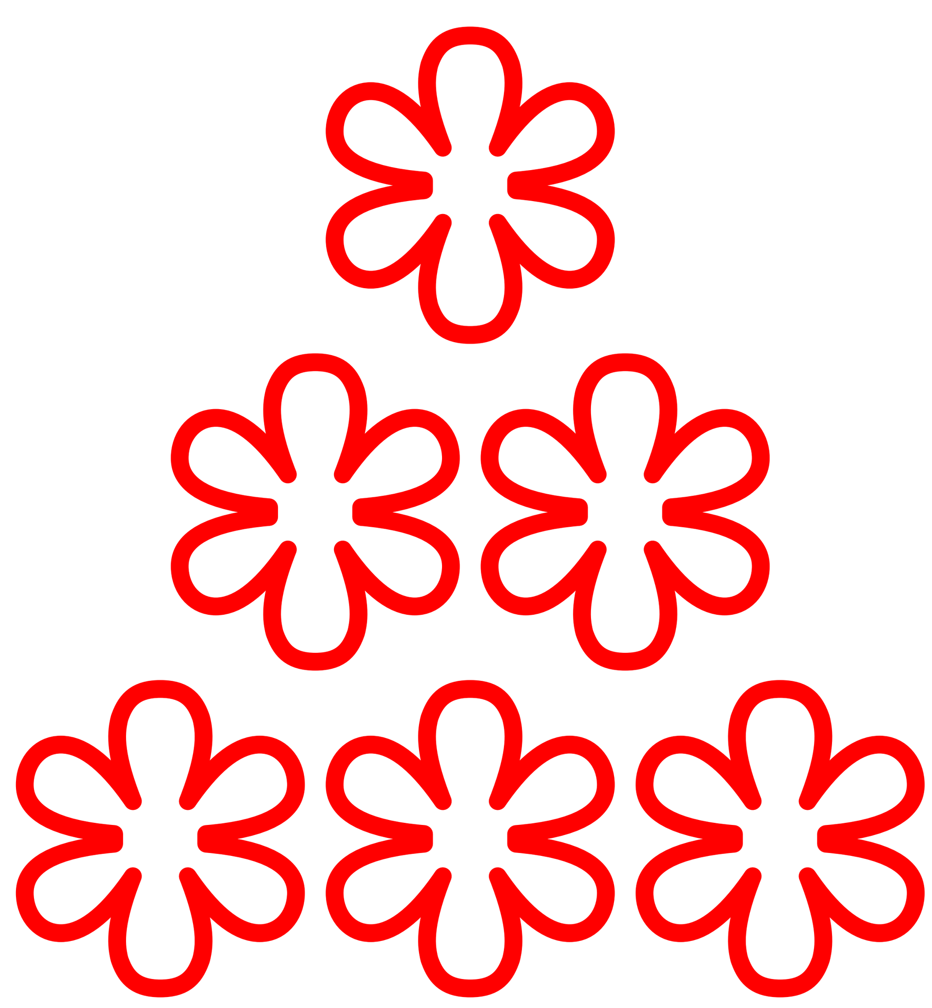
Listing 2: The tikz code for the almost perfect Michelin star 🤩



The `buggy Michelin star` generated with Claude



The `glitchy Michelin star` generated with a good old Python library



The `almost perfect Michelin star`

```
\label{lst:michelinmodel}
\newcommand\michelinman[2]{%
  \begin{tikzpicture}[baseline=-0.4ex]
    \foreach \i in {1,...,#2} {
      \begin{scope}[
        shift={(\i*5ex,0)},      % spacing between icons
        scale=0.8                % size adjustment
      ]
        \ifnum\i>#1
          % Empty (outline only)
          \draw[michelinred, line width=0.15pt]
            (6.24,0.69) -- (6.24,0.20)
            (6.24,0.20) .. controls (6.24,-0.54) and (5.77,-1.02) ..
               (5.05,-1.02)
            (5.05,-1.02) .. controls (4.27,-1.02) and (3.83,-0.53) ..
               (3.83,0.29)
            (3.83,0.29) -- (3.83,0.44)
            (3.83,0.44) .. controls (3.83,0.48) and (3.84,0.51) ..
               (3.84,0.51)
            (3.84,0.51) -- (3.85,0.68)
            (3.85,0.68) .. controls (3.11,0.15) and (2.52,0.15) ..
               (2.52,0.15)
            (2.52,0.15) .. controls (2.06,0.15) and (1.64,0.46) ..
               (1.64,0.46)
            (1.64,0.46) .. controls (1.24,0.75) and (1.05,1.22) ..
               (1.05,1.22)
            (1.05,1.22) .. controls (0.91,1.55) and (0.91,1.88) ..
               (0.91,1.88)
            (0.91,1.88) .. controls (0.91,2.89) and (2.07,3.42) ..
               (2.07,3.42)
            (2.07,3.42) -- (2.22,3.49)
            (2.22,3.49) .. controls (0.91,4.09) and (0.91,5.11) ..
               (0.91,5.11)
            (0.91,5.11) .. controls (0.91,5.80) and (1.39,6.32) ..
               (1.39,6.32)
            (1.39,6.32) .. controls (1.39,6.84) and (1.99,6.89) ..
               (1.99,6.89)
            (1.99,6.89) -- (2.12,6.79)
            (2.12,6.79) .. controls (2.10,6.99) and (2.10,7.27) ..
               (2.10,7.27)
            (2.10,7.27) .. controls (2.10,8.02) and (2.57,8.495) ..
               (2.57,8.495)
            (2.57,8.495) .. controls (2.57,8.97) and (3.36,8.97) ..
               (3.36,8.97)
            (3.36,8.97) .. controls (4.15,8.97) and (4.595,8.475) ..
               (4.595,8.475)
            (4.595,8.475) .. controls (5.03,7.98) and (5.03,7.58) ..
               (5.03,7.58)
            (5.03,7.58) -- (5.03,7.19)
            (5.03,7.19) .. controls (5.77,7.72) and (6.35,7.72) ..
               (6.35,7.72)
            (6.35,7.72) .. controls (7.00,7.72) and (7.475,7.195) ..
               (7.475,7.195)
            (7.475,7.195) .. controls (7.95,6.67) and (7.95,5.99) ..
               (7.95,5.99)
            (7.95,5.99) .. controls (7.95,4.97) and (6.79,4.44) ..
               (6.79,4.44)
            (6.79,4.44) -- (6.64,4.37)
            (6.64,4.37) .. controls (7.95,3.77) and (7.95,2.76) ..
               (7.95,2.76)
            (7.95,2.76) .. controls (7.95,2.11) and (7.48,1.57) ..
               (7.48,1.57)
            (7.48,1.57) .. controls (7.48,1.03) and (6.345,1.03) ..
               (6.345,1.03)
            (6.345,1.03) .. controls (5.175,1.45) and (5.175,1.45) ..
               (5.175,1.45)
            (5.175,1.45) -- (5.40,-1.34)
            (5.40,-1.34) .. controls (6.56,-3.03) and (7.55,-3.03) ..
               (7.55,-3.03)
            (7.55,-3.03) .. controls (7.96,-3.03) and (8.295,-2.66) ..
               (8.295,-2.66)
            (8.295,-2.66) .. controls (8.63,-2.29) and (8.63,-2.20) ..
               (8.63,-2.20)
            (8.63,-2.20) .. controls (8.63,-0.96) and (5.95,-0.73) ..
               (5.95,-0.73)
            (5.95,-0.73) -- (5.95,-0.44)
            (5.95,-0.44) .. controls (7.28,-0.33) and (7.955,0.04) ..
               (7.955,0.04)
            (7.955,0.04) .. controls (8.63,0.41) and (8.63,1.03) ..
               (8.63,1.03)
            (8.63,1.03) .. controls (8.63,1.48) and (8.305,1.855) ..
               (8.305,1.855)
            (8.305,1.855) .. controls (7.98,2.23) and (7.55,2.23) ..
               (7.55,2.23)
            (7.55,2.23) .. controls (6.54,2.23) and (5.40,0.53) ..
               (5.40,0.53)
            (5.40,0.53) -- (5.14,0.68)
            (5.14,0.68) .. controls (5.71,1.84) and (5.71,2.64) ..
               (5.71,2.64)
            (5.71,2.64) .. controls (5.71,3.88) and (4.58,3.88) ..
               (4.58,3.88)
            (4.58,3.88) .. controls (4.04,3.88) and (3.74,3.55) ..
               (3.74,3.55)
            (3.74,3.55) .. controls (3.45,3.20) and (3.45,2.67) ..
               (3.45,2.67)
            (3.45,2.67) .. controls (3.45,1.87) and (4.02,0.68) ..
               (4.02,0.68)
            (4.02,0.68) -- (3.76,0.54)
            (3.76,0.54) .. controls (2.62,2.24) and (1.61,2.24) ..
               (1.61,2.24)
            (1.61,2.24) .. controls (1.20,2.24) and (0.865,1.885) ..
               (0.865,1.885)
            (0.865,1.885) .. controls (0.53,1.53) and (0.53,1.06) ..
               (0.53,1.06)
            (0.53,1.06) .. controls (0.53,-0.18) and (3.20,-0.41) ..
               (3.20,-0.41)
            (3.20,-0.41) -- (3.20,-0.70)
            (3.20,-0.70) .. controls (0.53,-0.92) and (0.53,-2.17) ..
               (0.53,-2.17)
            (0.53,-2.17) .. controls (0.53,-2.61) and (0.85,-2.99) ..
               (0.85,-2.99)
            (0.85,-2.99) .. controls (1.17,-3.37) and (1.61,-3.37) ..
               (1.61,-3.37)
            (1.61,-3.37) .. controls (2.60,-3.37) and (3.76,-1.68) ..
               (3.76,-1.68)
            (3.76,-1.68) -- (4.02,-1.82)
            (4.02,-1.82) .. controls (3.45,-2.99) and (3.45,-3.82) ..
               (3.45,-3.82)
            (3.45,-3.82) .. controls (3.45,-4.35) and (3.74,-4.70) ..
               (3.74,-4.70)
            (3.74,-4.70) .. controls (4.04,-5.03) and (4.58,-5.03) ..
               (4.58,-5.03)
            (4.58,-5.03) .. controls (5.71,-5.03) and (5.71,-3.80) ..
               (5.71,-3.80)
            (5.71,-3.80) .. controls (5.71,-2.99) and (5.14,-1.82) ..
               (5.14,-1.82)
            -- cycle;
        \else
          % Filled
          \pic {michelinman=michelinred};
        \fi
      \end{scope}
    }
  \end{tikzpicture}%
}
```

Listing 3: The buggy tikz code towards a Michelin star 👎

credits

software libraries: tikz, latex, svg2tikz
catering: eugene’s son, café torre et fraction, le petit alep, mos
soundtrack: hania rani, brian eno, santana, d’angelo
thank you tristan miller for reproducing “Persecution of New Ideas” by C. L. Blood
thank you friends: deepika tiwari, alireza nasirianfar, roxana pena mendieta, nadia gonzalez fernandez, lena mk, martin monperrus

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